

1.3.1 Networks and topologies

- ☐ Types of network:
 - LAN (Local Area Network)
 - WAN (Wide Area Network)
- ☐ Factors that affect the performance of networks
- ☐ The different roles of computers in a client-server and a peer-to-peer network
- ☐ The hardware needed to connect stand-alone computers into a Local Area Network:
 - Wireless access points
 - Routers
 - Switches
 - NIC (Network Interface Controller/Card)
 - Transmission media
- ☐ The Internet as a worldwide collection of computer networks:
 - DNS (Domain Name Server)
 - Hosting
 - The Cloud
 - Web servers and clients
- ☐ Star and Mesh network topologies

Required

- ✓ The characteristics of LANs and WANs including common examples of each
- ✓ Understanding of different factors that can affect the performance of a network, e.g.:
 - Number of devices connected
 - Bandwidth
- ✓ The tasks performed by each piece of hardware
- ✓ The concept of the Internet as a network of computer networks
- ✓ A Domain Name Service (DNS) is made up of multiple Domain Name Servers
- ✓ A DNS's role in the conversion of a URL to an IP address
- ✓ Concept of servers providing services (e.g. Web server → Web pages, File server → file storage/retrieval)
- ✓ Concept of clients requesting/using services from a server
- ✓ The Cloud: remote service provision (e.g. storage, software, processing)
- ✓ Advantages and disadvantages of the Cloud
- ✓ Advantages and disadvantages of the Star and Mesh topologies
- ✓ Apply understanding of networks to a given scenario